DASFA-Net for Accurate Ship Detection in Optical Remote Sensing Images

Hanfu Li, Tianwei Zhang, Shikai Jiang, Xiyang Zhi, Dawei Wang, Lianru Gao, *Senior Member, IEEE*, Dong Yang, and Jindong Li

***Abstract*—Ship detection in optical remote sensing imagery is pivotal for various civilian and military applications. However, due to the presence of cluttered backgrounds and the signific ant variations in ship scales, accurately detecting ships in complex maritime environments remains challenging. To address these challenges, we propose the dual attention and scale-aware feature alignment network (DASFA-Net), to effectively detect multiscale ship targets under clutter interference. The DASFA-Net comprises two primary components: deformable spatial attention module with channel integration (DSAM-CI) and bidirectional flow alignment network (BiFAN). The DSAM-CI enhances discrimination between ships and distractors by jointly modeling correlations across the channel and spatial domains and applying deformable spatial attention. Concurrently, BiFAN employs a bidirectional feature flow alignment strategy to resolve spatial misalignment in feature fusion, significantly improving the quality of multiscale feature fusion. Lastly, we design the dynamic alpha complete intersection over union (DA-CIoU) based on prior knowledge of target characteristics to explicitly refine our detector for maritime ships. Extensive experiments on three challenging ship detection benchmarks, including HRSC2016, SCCOS, and FGSRCS have demonstrated the superiority of our DASFA-Net.**

***Index Terms*—Complex maritime scenes, feature alignment, multiscale feature fusion, optical remote sensing imagery, ship detection.**

I. Introduction

With the rapid growth of global maritime trade and the increasing demand for maritime security, accurate and efficient monitoring of ship targets at sea has become a critical remote sensing task in international maritime activities and marine resource management [1], [2]. Among available data sources, optical remote sensing has become the primary choice for ship detection owing to its high spatial resolution, intuitive interpretation, and wide coverage, supporting maritime traffic management, illegal-fishing enforcement, pollution monitoring, search-and-rescue, and global surveillance [3].

Meanwhile, data-driven deep learning approaches have become mainstream for optical ship detection [4]. Compared with traditional detection models that depend on handcrafted feature design [5], [6], deep learning methods automatically learn task-specific feature representations from large-scale optical remote-sensing datasets, substantially improving robustness to variations in target appearance, texture, and scale. Consequently, deep detection frameworks such as Faster R-CNN [7], YOLO [8], and SSD [9] have been widely adopted for object detection across diverse remote sensing scenarios.

However, due to the complexity of the maritime environment and the scale variability of ship targets, existing detection models still face the following two prominent challenges.

A. Complex Background Interference

Cluttered maritime backgrounds, such as wave patterns, cloud formations, islands, and dense port infrastructure, often mimic ship appearance or cause occlusion. This reduces separability between ships and distractors and increases false alarms and missed detections in complex scenes [10], [11], [12].

B. Diversity in Ship Scales and Appearances

Ship targets range from tiny few-pixel instances to image-spanning ships. Marked variations in target scale undermine the effectiveness of conventional CNN-based feature extractors, which are ill-equipped to capture and precisely localize objects that span a wide range of sizes [13], [14], [15].

Fig. 1 presents results of the representative detector RetinaNet. Background complexity leads RetinaNet to detect ship-like distractors as ships, producing false positives (shown in Fig. 1 complex background). In scenes with pronounced scale variation, it is difficult to accommodate targets across scales, causing missed detections (shown in Fig. 1 scale variation). These findings demonstrate that background interference and scale diversity substantially constrain the detection performance of conventional models in real-world maritime settings. Hence there is still an urgent need for more robust

Received 23 December 2025; revised 19 April 2026; accepted 5 May 2026. Date of publication 8 May 2026; date of current version 21 May 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62305086 and in part by the National Key Laboratory of Laser Spatial Information Foundation. *(Corresponding author: Shikai Jiang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3691594

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detectors. Methodologically, we summarize two mainstream remedies.

1) *Attention mechanisms to mitigate background interference.* Although works [16], [17] introduce channel and spatial attention to emphasize ship cues and suppress clutter weights. These methods [14], [18], [19] compute channel and spatial weights independently, preventing the capture of features that are simultaneously salient. Consequently, interference objects that are highly similar in either the channel or spatial domain are difficult to separate.
2) *Multiscale fusion to accommodate scale diversity.* Feature fusion networks [20], [21], [22] apply multilevel fusion to strengthen target information at different scales. However, existing methods [23], [24] tend to perform indiscriminate aggregation and place emphasis on feature enhancement at the fusion stage, which will result in feature misalignment and semantic mismatch [25]. Given that ships occupy limited spatial extent in remote sensing imagery, such spatial misalignments across resolutions more readily misassign targets to the background.

To address the problems caused by the above-mentioned defects, we aim to develop a multiscale ship detector in complex maritime scenarios, named as dual attention and scale-aware feature alignment network (DASFA-Net). To improve discrimination between ships and highly similar distractors, we propose the deformable spatial attention module with channel integration (DSAM-CI). DSAM-CI comprises two branches: channel attention and spatial attention. In the spatial branch, deformable convolution with an adaptive receptive field is utilized to align the sampling points with the shape, scale, and orientation of ships. This adaptation enables finer discrimination between ship targets and background clutter. Unlike conventional methods (e.g., CBAM [17], DANet [26]) that process spatial and channel attention independently, DSAM-CI adaptively fuses the two attention maps into a unified weight map. This design preserves more feature detail and captures higher-order interdependence between the channel and spatial domains. Consequently, the method amplifies weights of features that are jointly salient across both domains while suppressing distractors that are salient in only one, a behavior that is particularly beneficial in complex scenes. Furthermore, to resolve spatial misalignment between feature maps during feature fusion, we design the bidirectional flow alignment network (BiFAN). BiFAN adopts a bidirectional propagation structure to ensure thorough integration of multiscale features, which aligns feature maps through semantic flow, enabling precise matching between low-resolution semantic information and high-resolution details. Finally, by leveraging prior knowledge of target characteristics, we propose the dynamic alpha complete intersection over union (DA-CIoU) loss, which independently weights the IoU, localization loss, and aspect ratio loss to better adapt to ship detection tasks. Extensive evaluations on HRSC2016 and on SCCOS and FGSRCS, tailored to complex maritime environments, demonstrate that the proposed method significantly outperforms existing approaches.

The main contributions of this article are as follows.

1) We propose DASFA-Net, a novel ship detection framework tailored for multiscale target recognition in complex maritime environments. DASFA-Net achieves state-of-the-art results on three publicly available benchmark datasets.
2) We propose DSAM-CI to enhance discrimination between ships and distractors. By capturing correlations between channel and spatial domains and using deformable spatial attention, the module adaptively assigns higher weights to target features.
3) We design the BiFAN to correct spatial misalignment among feature maps. By leveraging bidirectional feature flow and a precise alignment mechanism, BiFAN learns pixel-wise offsets to accurately register deep semantics with shallow details.
4) We propose the DA-CIoU loss, which encodes ship-specific priors by independently weighting the IoU, center distance, and aspect ratio components, thereby specifically refining our detector for ship detection.

The rest of the article is organized as follows. Section II reviews related work. Section III provides a detailed description of our method and modules. In Section IV, experimental results are presented on three public datasets. Finally, Section V concludes the article and provides an outlook for future work.

II. Related Works

A. Ship Detection in Optical Remote Sensing Images

In optical remote sensing imagery, ship detection comprises two stages, namely traditional methods and deep learning methods. Together, these approaches satisfy the task needs in most routine maritime scenarios. Early optical ship detection studies predominantly relied on conventional methods. These approaches combined handcrafted features with prior knowledge, such as shape–background cues, saliency, and multilevel visual contrast [27], [28], [29]. However, these traditional methods have limited applicability and are suitable only for simple tasks. With the demonstrated success of deep learning in visual perception tasks, neural network-based methods have become the primary approach for ship detection. Achieving superior detection in complex scenes has become a central research focus in the field [30], [31], [32].

For example, MSRO-Net [33] employs a CNN-Transformer hybrid backbone and incorporates a coordinate-aware pyramid feature aggregation module to strengthen multiscale ship target representation. By performing feature decoupling and integrating a mask enhancement module for background suppression, DFDNet [34] alleviates missed detections and false alarms under complex conditions. To further improve detection efficiency, one-stage detectors have gained increasing adoption. Hu et al. [35] introduce context modulation and lightweight attention within the YOLO framework, improving robustness under low-visibility conditions and significantly reducing the false-alarm rate while maintaining inference efficiency. Ren et al. [36] utilized an FPN-based detection framework and employed saliency-guided feature fusion to augment the feature strength of moderately discriminative ship targets. Moreover, Yin et al. [14] introduced a high-order spatial interaction module along with a lightweight hybrid attention block

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(LHAB), thereby improving the robustness of small-scale ship detection while maintaining model efficiency. Recent studies such as KCPNet [37] and SMPISD-MTPNet [38] further indicate that context and scene modeling are important for improving ship detection robustness in complex environments. By controlling background clutter and scale variation, these studies outperform general-purpose detectors. Their performance, however, is unsatisfactory in highly complex scenes. We consider two primary causes: 1) discrimination between targets and highly similar distractors is inadequate; and 2) feature fusion lacks alignment across resolutions, which is particularly detrimental to remote sensing targets.

Accordingly, we propose a one-stage detection framework that makes significant progress in multiscale ship detection, particularly in complex scenes, while achieving an optimal balance between inference speed and localization accuracy.

B. Attention Mechanism

Originating in natural language processing (NLP) for sequence learning, attention allocates focus across sequence elements [39]. It has since been adopted in computer vision and remote sensing to emphasize target features and reduce background redundancy [40], [41]. In ship detection tasks, the target background often includes clouds, sea clutter, complex illumination, and interfering facilities. This inherent background complexity renders the integration of attention mechanisms particularly effective in substantially enhancing detection performance [1].

From the standpoint of implementation, attention modules applied to remote sensing ship detection can be classified into two categories: channel attention and spatial attention. Channel attention emphasizes discriminative feature channels by weighting each channel of the convolutional features. Examples include squeeze-and-excitation (SE) [16] and efficient channel attention (ECA) [42]. Spatial attention, on the other hand, explores the importance of different locations within the feature map along the spatial dimension, such as in the convolutional block attention module (CBAM) [17]. Researchers incorporate similar attention mechanisms into ship detection. These mechanisms direct the focus of the network toward key regions or channels that facilitate effective ship classification [43]. However, existing attention mechanisms exhibit limited adaptability to the diverse morphologies of ships and do not fully capture the deep interdependencies between channel and spatial information. This leads to two issues. First, the attention map often fails to coincide with the ship extent, resulting in incomplete target coverage or background leakage. Second, distractors that are similar in only one domain, either channel or spatial, are difficult to suppress.

The rise of transformer has further expanded the application boundaries of attention mechanisms. By utilizing self-attention to globally model image features, transformers have achieved breakthrough progress in object detection, also providing new perspectives for ship target detection in remote sensing images [44], [45]. Self-attention-based global modeling is computationally expensive, making it challenging to maintain real-time performance in practical applications. Although global attention mechanisms enhance holistic semantic representations, overreliance on global context can impair the extraction of localized, fine-grained features—an issue that is especially critical for ship targets, which occupy small image regions and exhibit significant shape variability. Consequently, detection frameworks that depend solely on self-attention are inherently limited in their performance on maritime object detection tasks.

To address these challenges, this article proposes DSAM-CI, which features a flexible receptive field and adaptively learns higher-order correlations between channel and spatial information.

C. Feature Fusion Network

By integrating high-level semantic information with low-level spatial details, feature fusion forms a multiscale feature pyramid whose levels jointly provide rich semantic context and high-resolution localization cues [46], [47]. This arrangement allocates targets of different scales to the most appropriate level, alleviating performance degradation arising from scale effects.

For example, FPN transfers deep semantic information to shallow feature maps through a top-down pathway, enabling both localization and classification capabilities at different scales [20]. However, during the top-down fusion process, FPN often suffers from loss of detailed information. To address this, PANet [48] introduces a bottom-up pathway, which further integrates shallow local details and deep semantic information, forming a bidirectional fusion structure. Subsequently, several studies have refined the fusion pathways using approaches such as NAS-FPN [49] and Recursive-FPN [50], iteratively optimizing the network architecture to achieve enhanced performance. Additionally, DFF-Net [51] employs bidirectional dense fusion and embeds attention modules to effectively integrate information across layers, thereby improving recognition of multiscale targets. Typical method like PSPNet [52] enhances the ability of network to perceive global background context through pyramid pooling in semantic segmentation tasks. However, most approaches typically ignore the impact of progressive sampling and context aggregation, which introduce spatial misalignment between feature maps at different resolutions. In remote sensing ship imagery where targets occupy limited spatial extent, the consequent ambiguity of semantic cues and imprecision of spatial details more severely impair detection performance.

To address these issues, the proposed BiFAN module performs precise alignment and complementary fusion of multiresolution features, minimizing detail loss during propagation and enabling full exploitation of multiscale information.

III. Proposed Method

A. Method Overview

To address the challenges of detecting maritime ship targets in complex scenarios, this article proposes the DASFA-Net framework, the overall structure of which is illustrated in Fig. 2. The framework consists of three main components: the backbone, BiFAN, and head. Specifically, the backbone adopts a structure similar to YOLOv8 [53], consisting of convolution layer (Conv), cross stage partial bottleneck with 2 convolutions (C2f), and spatial pyramid pooling–fast (SPPF).

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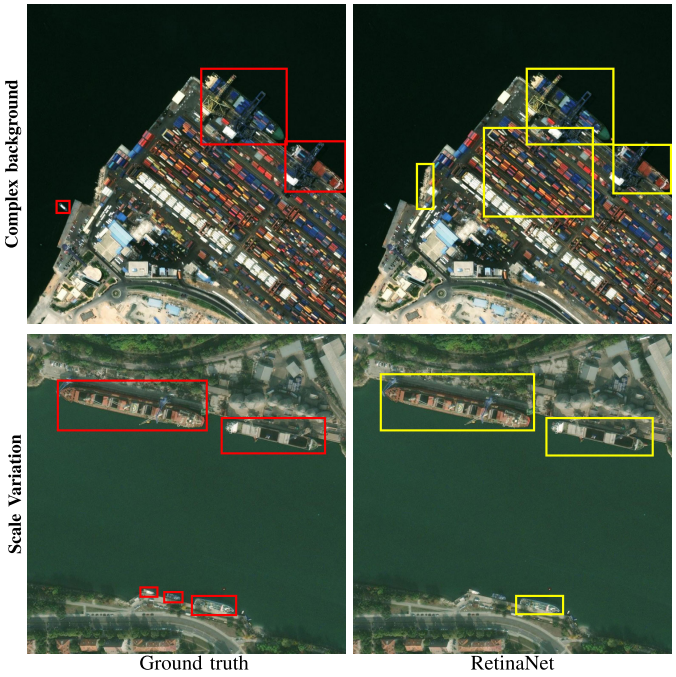


Fig. 1. Ground truth and RetinaNet detection results for representative scenes. Red bounding boxes denote ground truth; yellow bounding boxes denote detections.

Furthermore, DSAM-CI is incorporated into the backbone network to reweight feature maps for better distinguishing targets from distractors (Section III-B).

Three multilevel backbone feature maps are fused by BiFAN (Section III-C), which eliminates spatial misalignment to ensure high-quality integration of semantics and details. The resulting multiscale outputs are passed to the detection head for bounding-box localization and class prediction. We employ binary cross-entropy for classification and introduce the DA-CIoU loss for bounding-box regression to improve the network's adaptability to ship targets (Section III-D).

B. Deformable Spatial Attention Module With Channel Integration

DSAM-CI is an attention mechanism module embedded within the backbone to enhance feature extraction capabilities. This module decomposes input feature information into two independent branches: channel attention and spatial attention, each of which is processed separately with independent weighting. Finally, the independently weighted features from both branches are adaptively fused into a comprehensive attention map, which is then applied to the input features for fine-grained feature weighting. The structure of the DSAM-CI is illustrated in Fig. 3.

Specifically, given an intermediate feature map $\boldsymbol{F} \in \mathbb{R}^{C\times H\times W}$ as input, the channel attention mechanism first applies average pooling and max pooling operations along the spatial dimensions to aggregate global feature information. This process generates two channel-wise descriptors, $\boldsymbol{F}^{\mathrm{c}}_{\mathrm{avg}} \in \mathbb{R}^{C\times 1\times 1}$ and $\boldsymbol{F}^{\mathrm{c}}_{\mathrm{max}} \in \mathbb{R}^{C\times 1\times 1}$. These descriptors enable the model to capture both global contextual information and diverse feature representations, thereby improving the effectiveness of the attention mechanism while reducing the computational complexity of continuous calculations. Next, the two descriptors are passed through a shared network to generate the channel attention map $\boldsymbol{M}_{\mathrm{c}}(\boldsymbol{F}) \in \mathbb{R}^{C\times 1\times 1}$. This shared network consists of a multilayer perceptron (MLP) with one hidden layer. To minimize parameter redundancy, the hidden activation size is set to $\mathbb{R}^{C/r\times 1\times 1}$, where r represents the reduction ratio.

After obtaining the attention maps, an elementwise addition operation is applied to integrate and output the refined feature. Mathematically, the channel attention map $\boldsymbol{M}_{\mathrm{c}}(\boldsymbol{F})$ is computed as follows:

$$\begin{aligned}\boldsymbol{M}_{\mathrm{c}}(\boldsymbol{F}) &= \sigma\left(\mathrm{MLP}\left(\mathrm{AvgPool}(\boldsymbol{F})\right) + \mathrm{MLP}\left(\mathrm{MaxPool}(\boldsymbol{F})\right)\right)\\ &= \sigma\left(\boldsymbol{W_1}\left(\boldsymbol{W_0}\left(\boldsymbol{F}^{\mathrm{c}}_{\mathrm{avg}}\right)\right) + \boldsymbol{W_1}\left(\boldsymbol{W_0}\left(\boldsymbol{F}^{\mathrm{c}}_{\mathrm{max}}\right)\right)\right)\end{aligned} \tag{1}$$

where σ denotes the sigmoid function; $\boldsymbol{W_0} \in \mathbb{R}^{C/r\times C}$ and $\boldsymbol{W_1} \in \mathbb{R}^{C\times C/r}$. The ReLU activation function is applied immediately after $\boldsymbol{W_0}$.

From the spatial perspective, we first apply average pooling and max pooling operations along the channel axis, then concatenate the resulting feature maps to generate a spatial feature descriptor. Pooling along the channel axis has been proven effective in highlighting important regions.

On the concatenated feature descriptor, a deformable convolutional layer produces a spatial attention map $\mathbf{M}_{\mathrm{s}}(\mathbf{F}) \in \mathbb{R}^{1\times H\times W}$ that modulates feature responses. The key benefit is its learnable sampling offsets, which allow the convolution to adaptively concentrate sampling on target regions and suppress background interference. Moreover, deformable convolution endows the operator with an adaptive receptive field that conforms to object size and shape, thereby improving multiscale feature extraction. Exploiting this reduced reliance on fixed kernel dimensions, we employ a smaller kernel to capture fine-grained local details (e.g., edges and textures) while the learned offsets effectively enlarge the receptive field to incorporate richer contextual information.

A detailed description of this process is provided below. We employ two pooling operations to aggregate the spatial information of the feature map, generating two 2-D feature descriptors: $\boldsymbol{F}^{\mathrm{s}}_{\mathrm{avg}} \in \mathbb{R}^{1\times H\times W}$ and $\boldsymbol{F}^{\mathrm{s}}_{\mathrm{max}} \in \mathbb{R}^{1\times H\times W}$. The descriptors represent average pooling and max pooling features, respectively. These descriptors are then concatenated and passed through a 5×5 deformable convolution layer, generating our 2-D spatial attention map $\boldsymbol{M}_{\mathrm{s}}(\boldsymbol{F})$. Mathematically, this can be expressed as follows:

$$\begin{aligned}\boldsymbol{M}_{\mathrm{s}}(\boldsymbol{F}) &= \sigma\left(f_D^{5\times 5}\left(\left[\mathrm{AvgPool}(\boldsymbol{F})\right); \mathrm{MaxPool}(\boldsymbol{F})\right)\right]\right)\\ &= \sigma\left(f_D^{5\times 5}\left(\left[\boldsymbol{F}^{\mathrm{s}}_{\mathrm{avg}}; \boldsymbol{F}^{\mathrm{s}}_{\mathrm{max}}\right]\right)\right)\end{aligned} \tag{2}$$

where σ denotes the sigmoid function, and $f_D^{5\times 5}$ represents a deformable convolution operation with a 5×5 kernel.

The obtained channel attention map $\boldsymbol{M}_{\mathrm{c}} \in \mathbb{R}^{C\times 1\times 1}$ and spatial attention map $\boldsymbol{M}_{\mathrm{s}} \in \mathbb{R}^{1\times H\times W}$ encode global spatial features and channel-wise feature information, respectively. To effectively integrate these two information sources and learn their interdependencies, we transform $\boldsymbol{M}_{\mathrm{c}}(\boldsymbol{F})$ and $\boldsymbol{M}_{\mathrm{s}}(\boldsymbol{F})$ into feature vectors of the same dimension

$$\begin{aligned}\boldsymbol{M}'_{\mathrm{c}}(\boldsymbol{F}) &= \mathrm{vec}\left(\boldsymbol{M}_{\mathrm{c}}(\boldsymbol{F})\right) \in \mathbb{R}^{C\times 1}\\ \boldsymbol{M}'_{\mathrm{s}}(\boldsymbol{F}) &= \mathrm{vec}\left(\boldsymbol{M}_{\mathrm{s}}(\boldsymbol{F})\right) \in \mathbb{R}^{(H\times W)\times 1}.\end{aligned} \tag{3}$$

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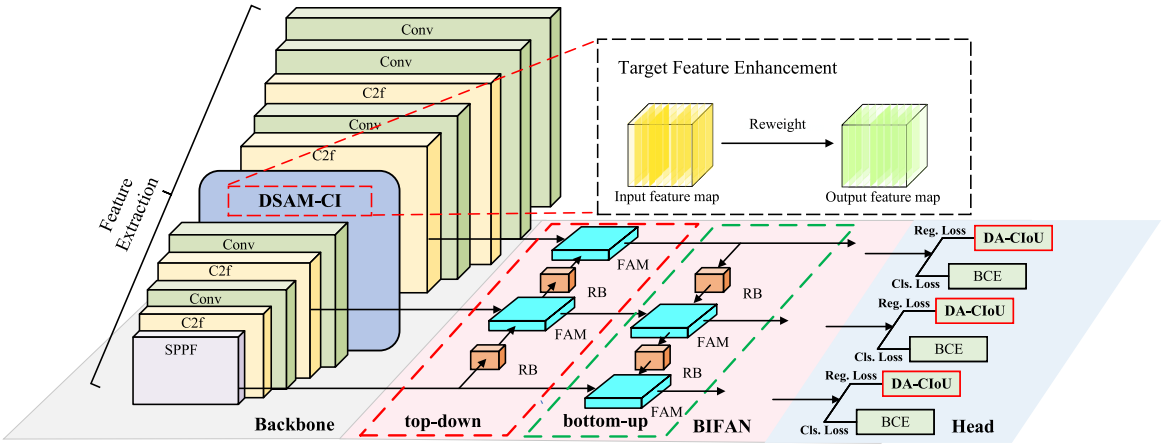


Fig. 2. Overview of the proposed framework. The top-down refers to the propagation of high-level semantic information from deeper to shallower layers, while the bottom-up refers to the propagation of local details from shallower to deeper layers. BCE denotes binary cross-entropy loss.

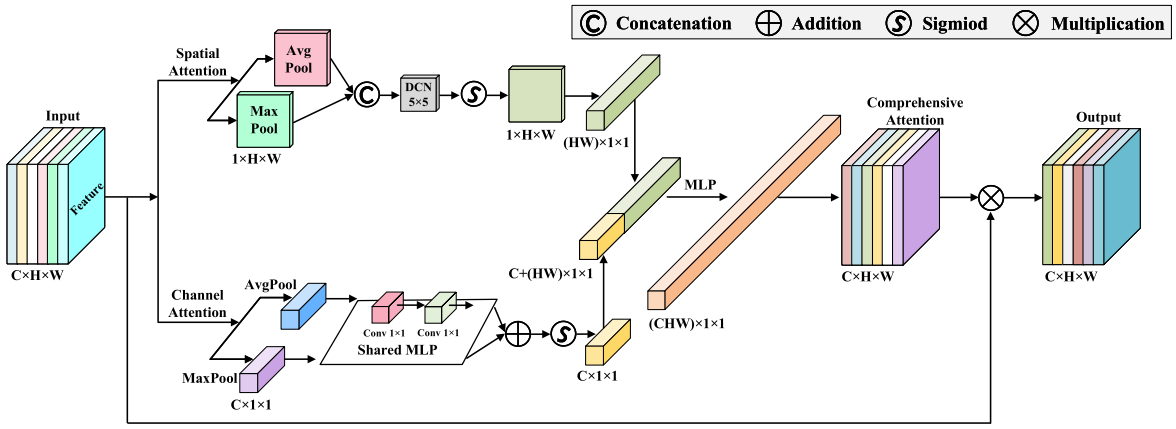


Fig. 3. Design of the DSAM-CI. DCN denotes deformable convolutional networks, and MLP denotes multilayer perceptron.

Then, we concatenate them to obtain a unified attention representation $\boldsymbol{M}_{\text{con}} \in \mathbb{R}^{(C+(H\times W))\times 1}$

$$\boldsymbol{M}_{\text{con}}(\boldsymbol{F}) = \text{Concat}\left(\boldsymbol{M}_{\text{c}}'(\boldsymbol{F}), \boldsymbol{M}_{\text{s}}'(\boldsymbol{F})\right) \in \mathbb{R}^{(C+(H\times W))\times 1}. \quad (4)$$

$\boldsymbol{M}_{\text{con}}$ is fed into an MLP with a single hidden layer to generate our final attention map $\boldsymbol{M}_{\text{mlp}} \in \mathbb{R}^{(C\times H\times W)\times 1}$. The activation size is set to $\mathbb{R}^{(C+(H\times W))/d\times 1}$ where d is the reduction ratio (set to 32 to balance model complexity and computational cost). This fully connected structure enables the model to capture higher-dimensional interactions between channel and spatial attention, improving robustness in complex scenarios.

After applying a reshape operation, the final attention map $\boldsymbol{M}_{\text{fin}} \in \mathbb{R}^{C\times H\times W}$ is obtained. The entire process can be summarized as follows:

$$\begin{aligned}\boldsymbol{M}_{\text{mlp}}(\boldsymbol{F}) &= \text{MLP}(\boldsymbol{M}_{\text{con}}(\boldsymbol{F})) \\ &= \boldsymbol{W_3}(\boldsymbol{W_2}(\boldsymbol{M}_{\text{con}}(\boldsymbol{F}))) \quad (5)\\ \boldsymbol{M}_{\text{fin}}(\boldsymbol{F})(c,h,w) &= \sigma\left(\boldsymbol{M}_{\text{mlp}}(\boldsymbol{F})[\,c\times(H\times W) \right. \\ &\quad \left. + (h\times W) + w\,]\right) \quad (6)\end{aligned}$$

where σ represents the sigmoid function, $\boldsymbol{W_2} \in \mathbb{R}^{(C+(H\times W))/d\times(C+(H\times W))}$, $\boldsymbol{W_3} \in \mathbb{R}^{(C\times H\times W)\times(C+(H\times W))/d}$. The ReLU activation function is applied immediately after $\boldsymbol{W_2}$.

The final refined feature map $\boldsymbol{F}'$ is obtained by performing an element-wise multiplication between the final attention map $\boldsymbol{M}_{\text{fin}}$ and the input feature map $\boldsymbol{F}$

$$\boldsymbol{F}' = \boldsymbol{M}_{\text{fin}}(\boldsymbol{F}) \otimes \boldsymbol{F} \quad (7)$$

where $\otimes$ denotes elementwise multiplication.

C. Bidirectional Flow Alignment Network

To effectively address the significant variations in the scale and shape of maritime ships, we design BiFAN as the Neck component of the network. BiFAN consists of multiple flow alignment modules (FAMs) [54] and residual blocks (RBs), forming a bidirectional feature flow that enables both top-down and bottom-up information propagation.

In the top-down pathway, low-resolution semantic information is transferred to high-resolution features to supplement global semantics, enhancing the contextual awareness of small-scale targets in complex backgrounds. In the bottom-up pathway, local details and edge information contained in

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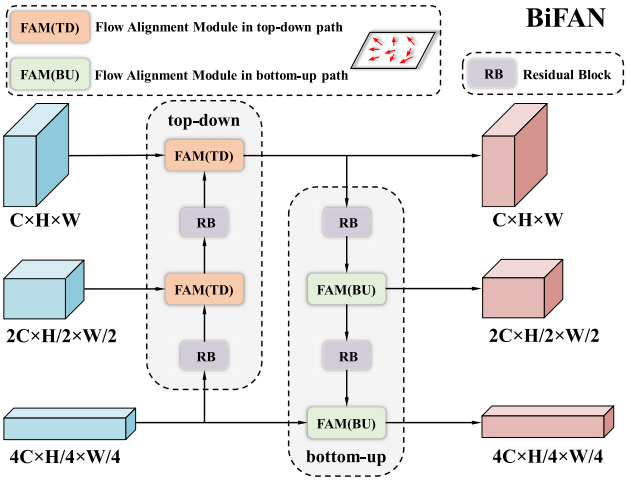


Fig. 4. Structure of BiFAN. BiFAN is a bidirectional feature alignment framework composed of FAM and RB modules. In the top-down path, high-level semantic information is propagated from low-resolution features to high-resolution features, while in the bottom-up path, local details are transmitted from high-resolution features to low-resolution features.

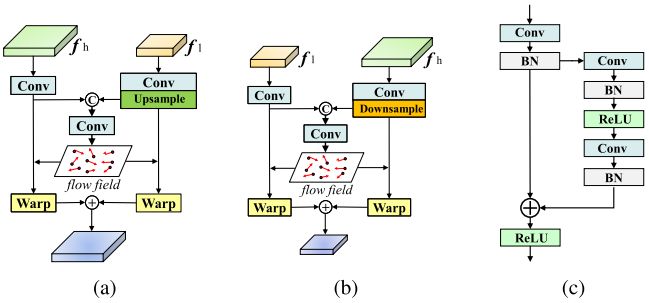


Fig. 5. Internal implementations of the FAM and RB used in BiFAN. (a) FAM in the top-down path. (b) FAM in the bottom-up path. (c) RB.

high-resolution features are transmitted back to low-resolution layers, refining the representation of large-scale targets and improving the overall scene depiction.

The overall structure of BiFAN is illustrated in Fig. 4. To improve multiscale feature fusion, we replace conventional 1×1 convolutions with RB modules for channel dimension adjustment, ensuring that multiscale features maintain consistent dimensions and statistical properties during feature propagation. Instead of the traditional sum operation, we introduce the FAM, which estimates and applies semantic flow to align feature maps of different resolutions. This approach alleviates spatial misalignment caused by conventional upsampling and downsampling, achieving precise semantic alignment.

Specifically, the RB module first adjusts the channel dimension of the input feature maps using a 1×1 convolution followed by batch normalization (BN), and then applies a residual connection to generate the output feature map. The FAM module takes multiscale feature maps as input and applies upsampling or downsampling as needed, depending on whether the current operation is in the top-down or bottom-up pathway. For a given high-resolution feature map f_h and low-resolution feature map f_l, FAM ensures that both are rescaled to the same spatial resolution and subsequently aligned by the learnable flow field, thereby facilitating effective feature fusion. Fig. 5 shows the specific structure of RB and FAM.

Once rescaled, the two feature maps are concatenated and passed through the flow field, which represents a flow or offset field that transfers semantic information between different feature levels. This mechanism describes the pixel-wise spatial transformation (alignment) from high-resolution to low-resolution feature maps. Warping is then applied based on the learned semantic flow to precisely align the feature maps, eliminating spatial misalignment, and ensuring accurate information propagation. Finally, FAM aggregates the aligned feature maps using a sum operation, integrating contextual information from both scales.

After completing feature alignment and fusion, BiFAN produces three output feature maps at different scales, which are subsequently fed into the detection head. By employing bidirectional feature interaction and precise alignment, semantic representations and spatial details are more effectively integrated, thus mitigating the semantic gaps and spatial misalignments typically introduced by conventional upsampling or naive feature concatenation. This design ultimately enhances both detection accuracy and robustness in multiscale ship detection tasks.

D. Dynamic Alpha Complete Intersection Over Union

Traditional one-stage networks typically adopt CIoU [55] as the loss function, which has demonstrated effective optimization in both localization regression and aspect ratio adjustment. However, for ship detection, where targets exhibit distinct structural characteristics, there remains room for further improvement. To enhance the performance of the loss function, we integrate the Alpha IoU [56] concept with CIoU, proposing an improved loss function called DA-CIoU. By introducing independent adjustment parameters α_i, the loss function gains greater flexibility, improving the model's adaptability and generalization capability.

The DA-CIoU loss function is defined as follows:

$$L_{\text{DA-CIoU}} = 1 - \text{IoU}^{\alpha_1} + \left(\frac{\rho^2\left(B, B_{gt}\right)}{c^2}\right)^{\alpha_2} + (\beta v)^{\alpha_3} \tag{8}$$

$$\beta = \frac{v}{1 - \text{IoU} + v} \tag{9}$$

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w_g}{h_g} - \arctan\frac{w_p}{h_p}\right)^2 \tag{10}$$

where $\rho^2(B, B_{gt})$ represents the Euclidean distance between the center points of the predicted and ground-truth bounding boxes, and c denotes the diagonal length of the smallest enclosing box. Here, w_g, h_g represent the width and height of the ground-truth box, while w_p, h_p denote the width and height of the predicted box, respectively.

Alpha IoU introduces a weighting mechanism into the traditional IoU computation, allowing the loss function to be dynamically adjusted based on specific task requirements. For ship detection tasks, the importance of IoU, localization loss, and aspect ratio loss in CIoU is not entirely consistent. By introducing independent weighting factors for each component, the loss function can better adapt to the specific

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characteristics of ship detection, improving both accuracy and adaptability.

By adjusting α_i values, the model can emphasize different factors. When $\alpha_i > 1$, more attention is given to penalty terms, increasing the impact of the corresponding loss component. When $\alpha_i < 1$, the weight of the respective component is reduced, helping to improve overall object detection performance.

For remote sensing ship detection, where ships are densely arranged in areas such as harbors, IoU is critical in directly reflecting the overlap between predicted and ground-truth boxes. Thus, α_1 (i.e., IoU term) should be assigned a higher weight. Similarly, α_2 (i.e., localization loss term) should be increased, as ship targets in dense layouts may have overlapping center points or even fully overlapping bounding boxes, making it difficult for IoU alone to distinguish between different bounding boxes. The inclusion of center distance loss helps the model accurately locate each target, improving multiobject discrimination.

Finally, although ships exhibit relatively similar shapes, their bounding box aspect ratios can undergo significant variations due to different viewing angles and orientations in remote sensing images. Therefore, α_3 (i.e., aspect ratio loss term) should be carefully controlled to enhance model flexibility.

Experimental results demonstrate that for remote sensing ship detection tasks, setting $\alpha_1 = 2$, $\alpha_2 = 2$, and $\alpha_3 = 1$ achieves the best detection performance.

IV. Experiments

A. Datasets

To evaluate the proposed method, we conducted experiments on three optical remote sensing ship detection datasets: the widely used HRSC2016 [57], as well as the SCCOS [58] and FGSRCS [59] datasets, which are specifically designed for ship detection in complex optical remote sensing images.

1) *HRSC2016:* We utilized the High-Resolution Ship Collection 2016 (HRSC2016) dataset for our experiments. This dataset consists of 1061 images collected from six major ports and is divided into a training set (436 images), a validation set (181 images), and a test set (444 images). The image resolution ranges from 0.4 to 2 m, with image sizes varying from 300×300 pixels to 1500×900 pixels, covering diverse scenes and target characteristics. The dataset provides three levels of annotations, including ship category, ship type, and specific ship models. Since our study focuses on evaluating the performance of ship category detection, we use the first-level annotations in our experiments.
2) *SCCOS:* We also evaluated our method on the SCCOS ship detection dataset, which contains 4639 images and 19 078 annotated ship instances. This dataset integrates imagery from multiple platforms, including Google Earth, Microsoft Map, WorldView-3, Pleiades, Ikonos, OrbView-3, and Jilin-1 satellites, ensuring a high level of data diversity. The ship instance areas range from 40.10 to 255305.31 px^2, illustrating extreme scale variation. Additionally, SCCOS is designed to reflect real-world complex application environments, incorporating various challenging factors such as thin clouds, light haze, dense clouds, weak shadows, ocean clutter, and port infrastructure. This dataset allows us to test the robustness of our algorithm under complex environmental interferences.
3) *FGSRCS:* We further evaluated our method on the Fine-Grained Recognition for Oriented Ship against Complex Scenes (FGSRCS) dataset. It comprises 1066 optical images with 6127 annotated ship instances; image sizes range from 600 to 1300 pixels with spatial resolutions of 0.5–2.5 m. The dataset is split into 523/223/320 images for training/validation/testing, respectively. Images were collected from Google Earth across multiple countries with acquisition dates spanning 2001–2020. It contains 17 fine-grained ship categories, and every instance—regardless of size—is labeled with a five-parameter oriented bounding box (x, y, w, h, θ) suitable for rotation-aware detection. The scenes span both nearshore and offshore settings, and include cluttered, dense, small-target, and large scale-variation scenarios, enabling rigorous evaluation under complex conditions.

B. Experimental Settings

1) Implementation Details: The proposed network is implemented based on the PyTorch framework and trained end-to-end on an NVIDIA Quadro RTX 8000 GPU. The input image size is set to 800×800 pixels, to balance small object detection and computational efficiency.

We adopt the same training strategy for the three datasets. During training, we use the AdamW optimizer with a weight decay parameter of 0.0001 to update network weights after each iteration. The initial learning rate is set to 0.001 and dynamically adjusted using the Cosine Annealing strategy. In the first 10 epochs, a warm-up strategy is applied, gradually increasing the learning rate from 0.000001 to 0.001, after which it is gradually reduced to 0.000001. Our proposed method as well as the comparative approaches are trained for a total of 300 epochs, with a batch size of 8.

After multiple comparative experiments, we replace the CIoU loss function with the proposed DA-CIoU loss for bounding box regression. Empirical results show that setting $\alpha_1 = 2$, $\alpha_2 = 2$, and $\alpha_3 = 1$ yields the best performance.

For HRSC2016, we follow the original dataset split for our experiments. For SCCOS, we select 4,353 RGB image samples and split the dataset into training (70%), validation (20%), and test (10%) sets. Additionally, since SCCOS only provides rotated bounding box annotations, we convert the minimum enclosing rectangles into their corresponding horizontal bounding-box representations for compatibility with our detection framework.

2) Evaluation Metric: We use Precision (P), Recall (R), $F1$-score, Average Precision (AP), false alarm rate (FAR), and frames per second (FPS) as evaluation metrics for the proposed model. P, R, $F1$, and AP are positively correlated with model performance.

Precision (P) is defined as the percentage of correctly predicted targets among all predictions, which measures the accuracy of detection. Recall (R) quantifies the missed detection rate, representing the percentage of correctly predicted

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TABLE I
ABLATION STUDIES ON HRSC2016 DATASET

DSAM-CI	BiFAN	DA-CIoU	P (%)	R (%)	F1	AP50 (%)	AP75 (%)	AP (%)	Params	GFLOPs
×	×	×	90.52	74.54	0.816	89.88	69.04	67.55	25.9 M	123.8 G
✓	×	×	93.76	77.66	0.848	92.96	72.56	70.54	26.2 M	124.3 G
×	✓	×	91.53	78.72	0.846	92.14	71.13	69.67	26.7 M	126.1 G
×	×	✓	91.74	74.33	0.821	91.06	70.65	68.32	25.9 M	123.8 G
✓	✓	×	93.55	81.58	0.870	94.10	74.08	72.49	27.0 M	126.6 G
✓	✓	✓	**93.88**	**82.07**	**0.875**	**94.53**	**75.28**	**73.58**	27.0 M	126.6 G

Note: Best results are in bold.

targets among all labeled targets. Their definitions are as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{11}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{12}$$

where TP denotes true positives, FP denotes false positives, and FN denotes false negatives.

The $F1$-score is defined as the harmonic mean of precision and recall

$$F1 = \frac{2PR}{P + R}. \tag{13}$$

AP is computed as the area under the precision–recall (PR) curve

$$\mathrm{AP} = \int_0^1 P(R)\, dR. \tag{14}$$

In ship detection tasks, especially in complex scenarios, the false detection probability directly determines the practical applicability of the proposed algorithm. Therefore, we introduce the FAR as a key metric, which is defined as follows:

$$\mathrm{FAR} = \frac{N_{fa}}{N_{DC}} \tag{15}$$

where N_{fa} represents the number of false detections, and N_{DC} represents the total number of predicted targets. A lower FAR value indicates better performance.

Parameters (Params) and floating point operations (FLOPs) are chosen to evaluate the computation complexity of models. The unit of Params is million, and the unit of FLOPs is GFLOPs (1 GFLOPs = 10^9 FLOPs). Frames/s (FPS) can assess the real-time performance.

C. Ablation Experiments

To evaluate the contribution of each proposed module, we conducted ablation experiments on two datasets, HRSC2016 and SCCOS. By comparing detection performance with and without these modules, we analyzed their impact on the final results. To ensure a fair comparison, all experiments were evaluated using the PASCAL VOC 2007 [60] protocol.

1) Effectiveness of Proposed Modules on HRSC2016: Table I presents the ablation results on the HRSC2016 dataset. The baseline achieved an AP of 67.55%, with AP at IoU = 0.5 (AP50) and AP at IoU = 0.75 (AP75) reaching 89.88% and 69.04%, respectively. The results indicate that each proposed module contributes significantly to improving detection performance. Incorporating the DSAM-CI led to an AP improvement of 2.99%, with AP50 and AP75 increasing by 3.08% and 3.52%, respectively. This result demonstrates the effectiveness of DSAM-CI in suppressing background interference and enhancing target feature representation. The BiFAN module, designed to strengthen multiscale feature fusion, improved recall by 4.18%, with AP increasing by 2.12%, AP50 by 2.26%, and AP75 by 2.09%, highlighting its adaptability to complex targets. The DA-CIoU module, which optimizes bounding box regression, resulted in an AP50 improvement of 1.18% and an AP75 increase of 1.61%. However, due to high-IoU matching optimization, the recall rate showed a slight decrease. When DSAM-CI and BiFAN were combined, the detection performance was further enhanced, leading to an AP improvement of 4.94%, with AP50 and AP75 increasing by 4.22% and 5.04%, respectively. The complete network, integrating all modules, achieved the best performance, with AP increasing by 6.03%, and AP50 and AP75 improving by 4.65% and 6.24%, respectively.

2) Effectiveness of Proposed Modules on SCCOS: Table II presents the ablation results on the SCCOS dataset, which contains more complex scenes compared to HRSC2016, posing greater challenges for the detection model. The baseline obtained an AP of 62.94%, with AP50 and AP75 reaching 86.72% and 64.51%, respectively. Compared to HRSC2016, the increased complexity of SCCOS resulted in a decrease in overall performance, yet the proposed modules still demonstrated significant improvements. The integration of the DSAM-CI module yielded substantial gains in challenging environments: AP increased by 3.60%, AP50 rose by 3.61%, and AP75 improved by 4.36%. Additionally, overall detection precision was enhanced by 3.80%, underscoring DSAM-CI's effectiveness in strengthening feature representations under complex background conditions. The BiFAN module, which facilitates multiscale feature fusion, resulted in a recall improvement of 5.08%, with AP increasing by 2.98%, AP50 by 2.35%, and AP75 by 2.83%, demonstrating its advantages in detecting small and complex targets. The DA-CIoU module further improved bounding box regression, leading to AP50 and AP75 improvements of 1.80% and 2.02%, respectively, while precision increased by 1.30%. However, the recall rate showed only a minor increase of 0.30%. When DSAM-CI and BiFAN were combined, AP improved by 5.97%, with AP50 and AP75 increasing by 4.86% and 5.57%, while recall increased by 7.76%.

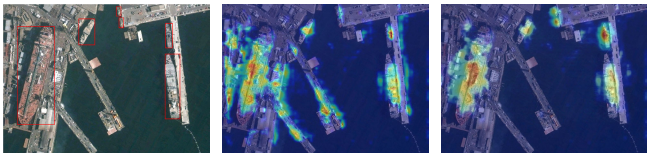
Ablation experiments on the HRSC2016 and SCCOS show that the proposed modules deliver substantial performance gains, with especially pronounced gains under

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TABLE II
RESULTS OF ABLATION EXPERIMENTS ON SCCOS DATASET

DSAM-CI	BiFAN	DA-CIoU	P (%)	R (%)	F1	AP50 (%)	AP75 (%)	AP (%)	Params	GFLOPs
×	×	×	86.85	70.06	0.776	86.72	64.51	62.94	25.9 M	123.8 G
✓	×	×	90.86	73.58	0.813	90.33	68.87	66.54	26.2 M	124.3 G
×	✓	×	88.55	75.14	0.812	89.07	67.34	65.92	26.7 M	126.1 G
×	×	✓	88.34	70.36	0.783	88.52	66.53	64.86	25.9 M	123.8 G
✓	✓	×	91.16	77.82	0.839	91.58	70.08	68.91	27.0 M	126.6 G
✓	✓	✓	**91.58**	**78.35**	**0.844**	**91.86**	**71.04**	**69.88**	27.0 M	126.6 G

Note: Best results are in bold.



(a) (b) (c)

Fig. 6. Visualization of DSAM-CI. (a) Ground truth. (b) Feature map before applying DSAM-CI. (c) Feature map after applying DSAM-CI.

TABLE III
RESULTS OF ABLATION EXPERIMENTS WITH DIFFERENT ATTENTION MECHANISMS

Dataset	Methods	P (%)	R (%)	F1 score	AP50 (%)
HRSC2016	SE	91.55	75.82	0.829	91.04
	CBAM	92.53	76.46	0.837	91.73
	CA	92.94	76.91	0.841	92.12
	DSAM-CI	**93.76**	**77.66**	**0.848**	**92.96**
SCCOS	SE	88.46	71.57	0.791	88.21
	CBAM	89.67	72.48	0.801	89.05
	CA	89.93	72.62	0.803	89.25
	DSAM-CI	**90.86**	**73.58**	**0.813**	**90.33**

Note: Best results are in bold.

extreme conditions. These improvements are achieved with only marginal changes in network parameters and FLOPs, indicating potential for operational deployment in real-world systems. DSAM-CI strengthens target feature representations while suppressing background clutter, thereby achieving a clearer separation between foreground and background and driving consistent gains across all evaluation metrics. BiFAN precisely aligns and fuses multiscale features, substantially enriching target representations. As a result, missed detections are reduced and recall is markedly improved. DA-CIoU refines bounding-box regression, yielding tighter boxes that better align with ship characteristics and therefore produces a marked increase in precision. When integrated, these components jointly address the two principal obstacles in optical ship detection—scale diversity and background clutter.

3) Comparisons With Different Attention Mechanisms: Table III presents the detection performance comparison on the HRSC2016 and SCCOS datasets when employing SE [16], CBAM [17], CA [61], and the proposed DSAM-CI. On HRSC2016, DSAM-CI achieves improvements of 0.007 in $F1$-score and 0.84 in AP50 compared to CA. In the SCCOS dataset, which contains more complex scene disturbances, DSAM-CI achieves 0.01 higher $F1$-score and 1.08% higher AP50 than CA. These results demonstrate that DSAM-CI, with its adaptive fine-grained weighting mechanism, provides significant advantages over conventional methods for ship detection under complex maritime conditions. To visually illustrate the effect of DSAM-CI on ship feature representation, Fig. 6 presents the corresponding feature-map visualizations. The original feature map exhibits high dispersed activations throughout the ship hull, quay edge structures, and water surface texture, obscuring true ship signals; upon integration of DSAM-CI, a continuous and pronounced heat concentration emerges along the ship hull, with markedly refined contour delineation, while activations associated with quay infrastructure and parallel wave patterns are substantially attenuated. This clearly demonstrates the ability of the module to suppress background clutter and accentuate target features, thereby achieving robust foreground–background discrimination.

TABLE IV
RESULTS OF ABLATION EXPERIMENTS WITH DIFFERENT FEATURE FUSION NETWORKS

Dataset	Methods	P (%)	R (%)	F1 score	AP50 (%)
HRSC2016	PANet	90.52	74.54	0.816	89.88
	NAS-PANet	90.93	75.82	0.828	90.79
	BiFPN	91.20	76.65	0.833	91.32
	BiFAN	**91.53**	**78.72**	**0.846**	**92.14**
SCCOS	PANet	86.85	70.06	0.776	86.72
	NAS-PANet	87.42	72.05	0.789	87.50
	BiFPN	87.91	72.94	0.797	88.04
	BiFAN	**88.55**	**75.14**	**0.812**	**89.07**

Note: Best results are in bold.

4) Comparisons With Different Feature Fusion Networks: Table IV further compares the detection performance of PANet [48], NAS-PANet [49], BiFPN [46], and the proposed BiFAN in terms of precision (P), recall (R), $F1$-score, and AP50 on the HRSC2016 and SCCOS datasets. The results show that BiFAN consistently achieves the best overall performance across both datasets. Notably, on the HRSC2016 dataset, BiFAN achieves a recall rate and AP50 that are 2.07% and 0.82% higher, respectively, than those of the second-ranked BiFPN. On the more complex SCCOS dataset, BiFAN exhibits a remarkable improvement of 2.2% in recall rate and 1.03% in AP50, demonstrating superior robustness. In contrast, although other multiscale feature fusion networks yield some performance gains, they remain hampered in ship detection by insufficient handling of feature misalignment. The experimental results demonstrate that BiFAN employs a bidirectional feature flow

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TABLE V
RESULTS OF COMPARATIVE EXPERIMENTS ON HRSC2016 DATASET. BEST RESULTS ARE IN BOLD

Category	Methods	AP50 (%)	FAR (%)	FPS
Two-stage	Faster R-CNN [7]	86.48	13.56	9.3
	Cascade R-CNN [62]	88.04	10.82	7.8
	Oriented R-CNN [63]	92.72	9.12	7.7
	RoI Transformer [64]	89.30	9.65	6.8
	MSRO-Net [33]	89.90	–	12.5
Anchor-free	FCOS [65]	87.63	15.67	11.6
	CentripetalNet [66]	90.41	15.42	10.4
	CenterNet [67]	89.92	13.67	15.3
	Oriented RepPoints [68]	91.35	11.87	10.5
	DFDNET [34]	91.70	–	–
One-stage	S2A-Net [69]	91.04	13.42	8.7
	RetinaNet [70]	89.44	12.78	10.1
	TPH-YOLOv5 [71]	90.76	8.89	24.7
	YOLOv8 [53]	89.88	9.48	**34.6**
	CM-YOLO [35]	92.31	11.62	25.8
	Ours	**94.53**	**6.12**	29.8

TABLE VI
RESULTS OF COMPARATIVE EXPERIMENTS ON SCCOS DATASET. BEST RESULTS ARE IN BOLD

Category	Methods	AP50 (%)	FAR (%)	FPS
Two-stage	Faster R-CNN [7]	84.52	15.36	7.2
	Cascade R-CNN [62]	85.85	12.50	6.5
	Oriented R-CNN [63]	89.38	11.71	6.1
	RoI Transformer [64]	86.43	12.32	5.7
	Sparse R-CNN [72]	86.15	13.25	9.6
Anchor-free	FCOS [65]	84.13	16.57	9.0
	CentripetalNet [66]	86.81	14.84	8.3
	CenterNet [67]	86.27	15.18	13.0
	Oriented RepPoints [68]	88.04	12.96	8.2
	RT-DETR [45]	87.82	13.05	22.5
One-stage	S2A-Net [69]	87.29	13.82	7.5
	RetinaNet [70]	84.05	14.74	8.2
	TPH-YOLOv5 [71]	88.91	12.83	21.2
	YOLOv8 [53]	86.72	13.15	**30.1**
	CM-YOLO [35]	89.84	12.13	23.3
	Ours	**91.86**	**8.42**	26.2

and precise alignment strategy, substantially enhancing multiscale target detection. This design significantly reduces missed detections and provides a pronounced improvement in recall relative to conventional fusion networks.

D. Comparative Experiments

To verify the effectiveness and robustness of the proposed method, we compared our approach with several state-of-the-art methods, including one-stage detectors, two-stage detectors, and anchor-free detectors. The detection performance was evaluated on both the HRSC2016 and SCCOS datasets.

1) Results on HRSC2016: On the HRSC2016 dataset, the primary challenges arise from the complexity of port infrastructure and densely distributed ships. Table V presents the comparison results between our network and several advanced methods. Compared to classical algorithms such as Faster R-CNN, FCOS, and RetinaNet, the proposed approach achieves a notable improvement in AP50. When compared with Oriented R-CNN, Oriented RepPoints, and S2A-Net, which have demonstrated strong performance for ship detection, our method still achieves the highest AP50 value of 94.53%, surpassing Oriented R-CNN by 1.81%. Notably, our algorithm also attains the lowest FAR among all competing methods, reducing false alarms by 2.77% compared to TPH-YOLOv5, highlighting its effectiveness in controlling false detections.

In addition, given the inherent speed advantages of one-stage detectors, our method maintains a balance between detection accuracy and inference speed, achieving the second-highest FPS (29.8) among all methods while maintaining its performance advantages. Although YOLOv8 achieves a higher FPS, it falls 4.65% lower in AP50 and 3.36% higher in FAR, indicating a noticeable gap in detection capability.

We select the two-stage network Cascade R-CNN, the anchor-free network FCOS, and the one-stage network RetinaNet for visualization of detection results and compare them with the proposed method. As shown in Fig. 7, we choose scenarios with significant ship scale variations, dense distributions, and similar false alarm sources to comprehensively demonstrate detection performance.

It can be concluded that Cascade R-CNN achieves superior detection performance on large-scale targets through multistage regression; however, some small targets may be prematurely filtered out during iterations, leading to missed detections. FCOS, leveraging pixel-level centerness for anchor-free detection, is prone to false positives in homogeneous textures or highly dense environments. RetinaNet mitigates class imbalance via Focal Loss, yet its fixed anchor settings may still result in missed or false detections when ship scales and orientations vary. In contrast, the proposed method integrates multiscale attention and a fine-grained feature alignment strategy, enabling stable extraction of key ship contours even in complex harbor environments and multiscale ship scenarios. This significantly reduces both false alarms and missed detections.

2) Results on SCCOS: On the SCCOS dataset, which includes thin clouds, light haze, thick clouds, lighting variations, ocean clutter, and port structures, the results presented in Table VI further demonstrate the robustness of our model in complex maritime environments. The proposed method achieves an AP50 of 91.86%, outperforming the second-best CM-YOLO by 2.02%. Additionally, our method achieves the lowest FAR, reducing false alarms by 3.29% compared to Oriented R-CNN. In terms of inference speed, our method runs 3.6 times faster than Oriented R-CNN, demonstrating significant advantages for real-time detection applications.

Regarding detection speed, our method maintains the second-best FPS performance on the SCCOS dataset, trailing only YOLOv8. However, the AP50 improvement of 5.14% and FAR reduction of 4.73% on SCCOS surpass the improvements observed on HRSC2016, further confirming the model's superior adaptability to complex scenarios.

As shown in Fig. 8, in scenarios with significant cloud occlusion and strong interference from false alarm sources,

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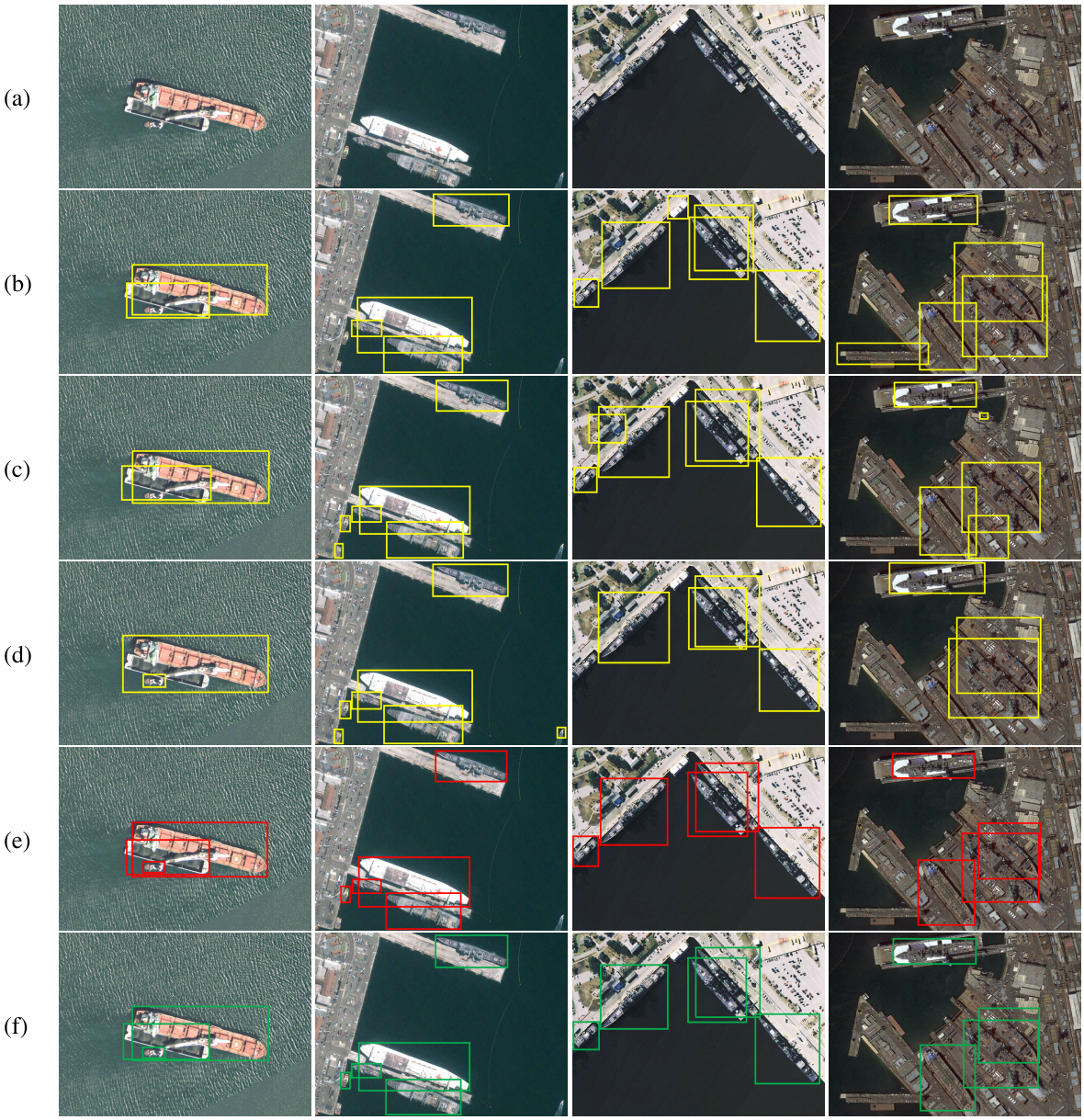


Fig. 7. Representative scenes with pronounced scale variation on the HRSC2016 dataset, with results from different considered methods. (a) Test image. (b) Cascade R-CNN. (c) FCOS. (d) RetinaNet. (e) Proposed method. (f) Ground truth.

Cascade R-CNN demonstrates superior detection performance compared to FCOS. However, under severe occlusion or highly similar background clutter, it still suffers from missed and false detections. RetinaNet exhibits a higher number of missed detections and lower regression accuracy. In contrast, the proposed method effectively captures key ship details in complex maritime backgrounds, enabling robust extraction of occluded ships while distinguishing highly similar interference sources. This ensures that the predicted bounding boxes are more closely aligned with the ground truth.

Evaluation on the two benchmarks demonstrates that the proposed network effectively alleviates the two core challenges in ship detection—multiscale targets and complex background interference. Furthermore, the network maintains robust detection performance while achieving processing speeds superior to those of most single-stage methods. This balance of efficiency and accuracy underscores its practical utility and strong adaptability under challenging operational conditions.

E. Extended Experiments

To further validate the robustness and generalization capability of the proposed method under highly challenging operational conditions, we conducted additional experiments on the fine-grained ship recognition in complex scenarios (FGSRCS) dataset. This dataset is more difficult than HRSC2016 and SCCOS since it embeds heavy environmental interference and upgrades the task to fine-grained recognition across 17 ship categories. Therefore, evaluation on FGSRCS

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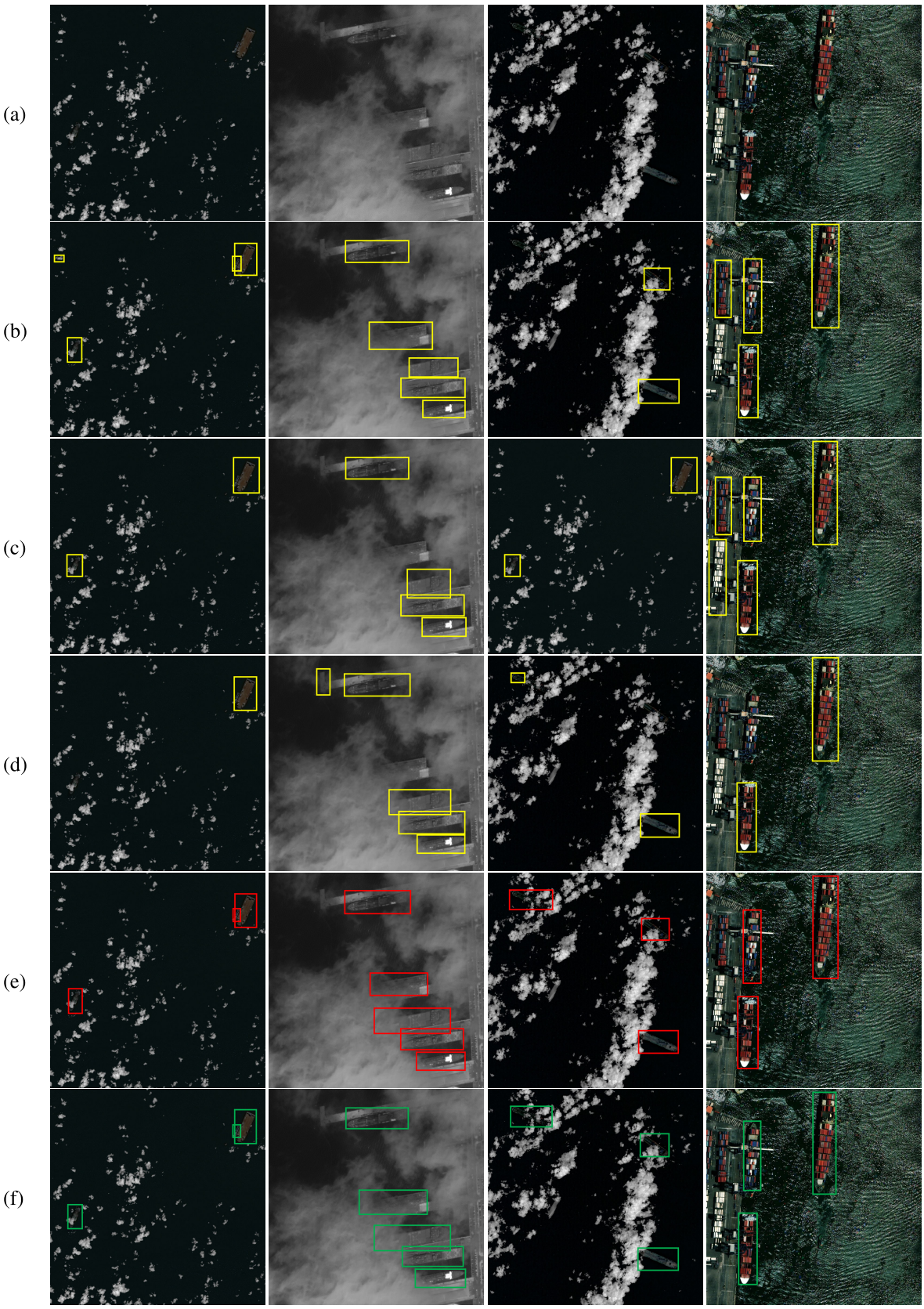


Fig. 8. Representative scenes with pronounced complex background on the SCCOS dataset, with results from different considered methods. (a) Test image. (b) Cascade R-CNN. (c) FCOS. (d) RetinaNet. (e) Proposed method. (f) Ground truth.

provides a rigorous test of the ability of DASFA-Net to extract highly discriminative features in cluttered environments.

Table VII shows that DASFA-Net attains 84.7% AP50 and 81.8% AP75, outperforming all strong baselines. Extended

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TABLE VII
SHIP DETECTION RESULTS ON FGSRCS DATASET. BEST RESULTS ARE IN BOLD

Method	mAP50 (%)	mAP75 (%)
RetinaNet [70]	47.8	35.4
S2ANet [69]	59.6	50.5
R3Det [73]	67.0	54.8
Faster RCNN [7]	63.8	30.3
EIRNet [51]	76.0	66.6
ROI Transformer [64]	76.7	73.3
Oriented RCNN [63]	79.3	72.8
ReDet [74]	81.9	79.3
SASM Reppoints [75]	53.3	45.6
ARS DETR [44]	70.4	64.8
Ours	**84.7**	**81.8**

experiments demonstrate that DASFA-Net achieves strong discrimination of nonship distractors and closely neighboring ship categories by emphasizing ship-hull features that are simultaneously salient in channel and spatial domains. This indicates large interclass margins under fine-grained ambiguity and further verifies its adaptability to complex scenes and multiscale targets, highlighting practical robustness without task-specific tuning.

V. CONCLUSION

In this article, we propose a novel DASFA-Net framework for accurate ship detection in optical remote sensing images. We identify that complex background interference and diversity in ship scales are two obstacles that hamper the ship detection task. To address these, we propose the DSAM-CI, an attention mechanism that adaptively fuses key channel and spatial cues to improve discriminative capability between ship targets and distractors. Afterward, the BiFAN is designed to eliminate the feature misregistration inherent in conventional fusion architectures, which enhances multiscale feature fusion. Furthermore, the DA-CIoU encodes ship-specific geometric priors by independently weighting regression components, thereby enhancing the adaptability of the model to ship targets. The experimental results demonstrate that DASFA-Net achieves outstanding performance on the HRSC2016, SCCOS, and FGSRCS datasets. In particular, the results on the challenging SCCOS and FGSRCS datasets highlight the strong adaptability of the framework to complex maritime scenarios.

Despite these advancements, the proposed method still leaves room for further improvement, especially in extreme conditions such as adverse weather or severe occlusion. Future work will focus on multimodal data fusion and lightweight network design to enhance real-time processing capability while maintaining high detection accuracy.

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